# CBDC-PUF: A Novel Physical Unclonable Function Design Framework Utilizing Configurable Butterfly Delay Chain Against Modeling Attack

YANJIANG LIU, JUNWEI LI, TONGZHOU QU, and ZIBIN DAI, Information Engineering University



Physical unclonable function (PUF) is a promising security-based primitive, which provides an extremely large number of responses for key generation and authentication applications. Various PUFs have been developed as central building blocks in cryptographic protocols and security architectures, however, the existing PUFs and their improvements are still vulnerable to modeling attacks (MA) with refined machine learning algorithms. In this article, a configurable butterfly delay chain-based PUF design framework is proposed to meet the requirements of randomness, reliability, uniqueness, and MA-resistance metrics. A configurable butterfly delay chain is introduced to create multiple pairs of symmetric paths and a strong PUF relying on the intrinsic delay fluctuations of two identical paths is built. Furthermore, a secure hash function is used to insert non-linearities into the PUF, and a BCH-based error correction algorithm is utilized to recover the actual responses under noisy environments. The proposed PUF is implemented on Xilinx FPGAs and three machine learning algorithms are used to evaluate the resistance against MA. Experimental results show that the randomness, reliability, and uniqueness of the proposed PUF are close to the ideal value (49.6%, 99.9%, and 49.9%, respectively), and the prediction accuracy reaches 50% that indicating a desirable resilient to MA.

CCS Concepts: • **Security and privacy → Security in hardware**; **Embedded systems security**;

Additional Key Words and Phrases: Physical unclonable function, modeling attack, machine learning algorithm, cryptographic hash function

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## 1 INTRODUCTION

Since chips are deployed in many critical applications and sensitive fields, their security, privacy, and safety aspects are of utmost concern. With developing numerous hardware attack techniques, chips are not enough to provide sufficient protection against these attack approaches that have become a serious threat to the security of systems. Although several security solutions have been

Authors' address: Y. Liu, J. Li, T. Qu, and Z. Dai (corresponding author), Information Engineering University, Shangcheng Road No. 12, Zhengzhou, Henan 450000, China; emails: liuyj_1013@126.com, lijunwei3@163.com, qutongzhou@outlook.com, daizb2004@126.com.

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explored in the literature, such as **Trusted Platform Modules** (**TPMs**) [1], those approaches introduce non-negligible area overhead, power consumption, and computational complexity, which make them difficult to be deployed in cost-sensitive systems. A promising approach to protect these systems against hardware attacks is physically unclonable functions. The main advantage of **Physical unclonable functions** (**PUFs**) is that no secret needs to be securely stored and the PUF response generation is simple. Therefore, PUFs are increasingly becoming the central building blocks in cryptographic protocols and security architectures.

Since the Arbiter-based PUF first presented by Lim et al. [2], PUF has attracted widely attention in the field of hardware security. After that, numerous PUFs, including the Arbiter-based PUF [3, 4], RO-based PUF [5–7], SRAM-based PUF [8, 9], Latch-based PUF [10, 11], Flash-based PUF [12, 13], and so on, have been proposed to strengthen the security of systems. Although various PUFs are explored over decades, only the butterfly PUF and SRAM PUF are widely used as low-cost security solutions for cryptographic products [14, 15]. Furthermore, several PUF-based security solutions, like the device identification and authentication [16], secure storage of cryptographic secret [17], IP protection [18], anti-counterfeiting [19], and so on, are already explored in the literature and integrated into the high-security products that circulate in the market. Among all existing PUFs, the butterfly PUFs achieve high reliability, randomness, and uniqueness, which have recently received increasing attention from both academics and the industry [20–22]. Therefore, the butterfly PUFs have become the most appealing option for establishing trust in resource-constrained embedded systems. However, the original butterfly PUF is a weak PUF and has a limited number of challenge–response pairs. In this case, the butterfly PUF is only used as the secret key, which does not scale to the authentication and attestation fields.

PUFs are based on the intrinsic physical differences caused by the random manufacturing process which are infeasible to predict and reproduce in practice. However, the PUF behavior with those differences is easily modeled by machine learning algorithms and the PUF's response of an arbitrary challenge can be predicted accurately by this mathematical model. Several works described in [3], [23], and [24] perform a successful **modeling attack** (**MA**) on the existing PUFs. Furthermore, several MA resistant PUFs, including the XOR **Arbiter-based Physical Unclonable Functions** (**APUFs**) [3, 25, 26], feedforward APUFs [2, 27, 28], lightweight secure PUFs [29], and so on, have been explored to resist the modeling attacks. Although several studies enhance the resistance against MA, they are still vulnerable under refined machine learning algorithms with enhanced power consumption and area cost. Moreover, PUFs are susceptible to the varying supply voltage and ambient temperature variations, and PUFs provide slightly different responses when queried with the same challenge multiple times, which cannot be used directly as cryptographic keys. In summary, the existing PUFs suffer from one or more of the following disadvantages: (1) they are vulnerable to modeling attacks with refined machine learning algorithms, (2) they are subject to varying operating conditions, and (3) they introduce non-negligible power consumption and area overhead to enhance the resistance against MA.

In this article, we design a high-security and low-overhead butterfly PUF to process security-critical and privacy-sensitive data. A path-configurable delay chain is designed to generate multiple combinations of two symmetrical paths, a cross-coupled circuit with two delay chains is constructed to produce two possible stable states and a flip-flop-based arbiter is used to settle to one of the two stable states. Then, the configurable delay chain of arbiter PUF and the nonlinear mathematical model of butterfly PUF are combined to form a **C**onfigurable **B**utterfly **D**elay **C**hain-based **P**hysical **U**nclonable **F**unction (**CBDC-PUF**), which produces a large number of **challenge/response pairs** (**CRPs**) and achieves a high MA-resistance ability. Furthermore, an MA-resilience solution based on the secure hash algorithm is introduced to break the linear relationship between the challenges and responses, which makes the conventional machine

learning algorithms hard to build the mathematical models of the proposed PUF. Besides, an error correction scheme based on the **Bose–Chaudhuri–Hocquenghen** (**BCH**) code is utilized to improve the robustness of PUF. The main contributions are listed as follows:

— A CBDC-PUF based on the unique differences of two symmetrical paths is proposed to produce a large number of CRPs, which can be easily integrated into the hardware security fields. There is no need for user to implement the power-up and power-down operations repeatedly, and the space size of CRPs is large enough to meet the requirement of cryptographic protocols.
— The butterfly PUF and a secure hash function are introduced to insert the non-linearity into the relationship between the challenges and responses for improving the MA-resistance. The prediction accuracy of CBDC-PUF is close to 50%, which shows that the MA are equivalent to random guessing.

The rest of this article is organized as follows. Section 2 introduces an overview of existing PUFs. Section 3 presents the structure, working principle, and security analysis of proposed PUF. Section 4 describes the experimental setup, analyzes the main performances and MA-resistance of proposed PUF. Section 5 concludes this article.

## 2 BACKGROUND

### 2.1 Attack Model

Recently, various third-party EDA tool vendors, Intellectual Property providers, and foundries from different countries and regions are participating in the PUF design and fabrication process. We assume that the total PUF design and fabrication stage are trusted, and its toolchain is reliable. Therefore, the attack scenario is mainly at the application stage, and adversaries can access the PUF-based chips but have little knowledge about the PUFs. Numerous attack approaches, including micro probing attack, side-channel attack, fault injection attack, MA, and so on, are exploited by adversaries to retrieve the PUF responses in this stage. The actual PUF does not require several specific memory cells to store the responses and cannot be accessed through the communication interfaces. To evaluate the performance of PUF effectively, we suppose that some CRPs can be collected freely via interfaces.

### 2.2 Physical Unclonable Functions

Considering the security requirements of embedded systems, numerous PUFs have been proposed over the past several decades. It can be broadly classified into (1) non-electronic PUFs, (2) analog electronic PUFs, (3) delay-based PUFs, and (4) memory-based PUFs. Among all the existing PUF instantiations, delay- and memory-based PUFs are the most appealing ones for the embedded systems that are used in critical applications and sensitive fields. From [30], arbiter PUF is first presented by Lim et al. and its efficacy is evaluated on the application-specific integrated circuits. However, its linear additive path delays are also vulnerable to MA. To further improve the MA-resistance ability, a feed-forward arbiter PUF is proposed to use the intermediate value of arbiters as the parts of the challenge [2]. But References [31] and [32] have successfully implemented the MA on the feed-forward arbiter PUF. Majzoobi et al. demonstrate that advanced modeling techniques can predict the response of the feed-forward arbiter PUF [5, 33]. After that, Pranesh et al. introduce an XOR APUF design scheme [23], Rashid et al. present a reconfigurable arbiter PUF [3], Nimesh et al. design a hybrid nonlinear PUF [24], and Durga et al. propose a multiplexer-based composition of arbiter PUF design approach [4] to thwart MA. Among them, XOR APUF is the most efficient approach to resist MA. Although XOR arbiter PUFs improve the computational complexity

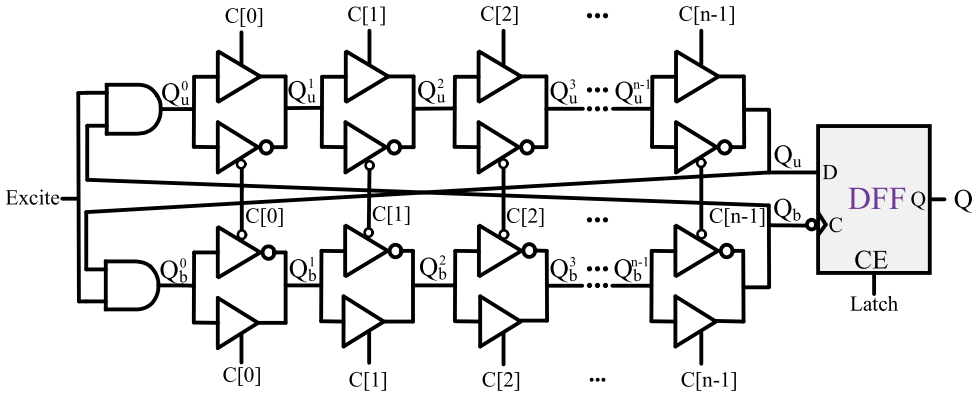



Fig. 1. Structure of CBDC-PUF.

of MA, the reliability decreases with the increasing number of XORs. Moreover, Ruhrmair et al. can model 16-XOR APUF using a combination of MA and side-channel attacks [34].

**Ring oscillator PUF** (**RO-PUF**) is another type of delay-based PUF, which exploit the differences among several ring oscillators to output an unpredictable response. However, the ring oscillator is susceptible to the fluctuations of voltage and ambient temperature, and the reliability of basic RO-PUF is relatively low. To improve the reliability of RO-PUF, numerous improved RO-PUFs, including the ARO-PUF [17], ACRO-PUF [35], CF-ROPUF [36], TERO-PUF [37], frequency offset-based RO-PUF [38, 39], and so on, have been developed to reduce the bit error rate of responses. Besides, References [6, 7, 40, 41] increase the number of CRPs of traditional RO-PUF. However, the tradeoff between the implementation cost, reliability, and MA-resistance is still the main challenge of the existing RO-PUFs.

Due to the existence of manufacturing process variations, the physical properties vary with each other. Memory cells have two possible stable states ("0" and "1"), the stable state a cell enters depends on the physical properties of its transistors. It is extremely difficult to guess the stable state of a specific cell. Memory-based PUFs exploit the instability behavior of the memory cells to produce physical unclonable and unpredictable responses. Since the SRAM-PUF first presented in [8], various memory cells, including the latches [10, 11], Flash [12, 13], D flip-flops [42, 43], and scan flip-flops [44–46], have been explored to design the memory-based PUF. Generally, the memory-based PUF is a weak PUF and the CRPs space size depends on the number of memory cells. The larger size the CRP space is, the more cells the memory-based PUF needs, and the larger the area occupies. Moreover, majority of memory-based PUFs require the power-up behavior to settle to a stable state from two possible states. There is no doubt that such memory-based PUFs need a dedicated power network to implement multiple power-up operations. In addition, the stable state stored in the memory cells can be revealed by various physical attacks (e.g., micro probing attack [47], side-channel attack [48], and fault injection attack [49, 50]), which makes the memory-based PUFs susceptible to physical attacks.

## 3 CONFIGURABLE BUTTERFLY DELAY CHAIN-BASED PUF DESIGN AND SECURITY ANALYSIS

### 3.1 PUF Design

Figure 1 shows the structure of CBDC-PUF. As shown in Figure 1, a cross-coupled circuit with two configurable delay chains is used to produce two possible states, and a **D-type flip-flop** (**DFF**) at the end of the cross-coupled circuit is used to capture the possible stable state. The configurable

delay chain is composed of $n$ configurable delay elements and 1 AND gate. Where *Excite* is an input signal of AND gate and *Latch* is the clock enable signal of DFF. Each configurable delay element includes a buffer and inverter, which provides two optional signal paths according to the value of control signal. The $i$th configurable delay element is used as an example to explain its structure. Where $C[i]$ and its complementary value ($\overline{C[i]}$) are the control signal of configurable delay element, $Q_{\mathrm{u}}^{i}$ and $Q_{\mathrm{u}}^{i+1}$ are the input and output signal of the $i$th configurable delay element. The logical value of $Q_{\mathrm{u}}^{i}$ propagates to the output $Q_{\mathrm{u}}^{i+1}$ when the value of input $C[i]$ is logic "0", while the opposite value of $Q_{\mathrm{u}}^{i}$ transmits to the output $Q_{\mathrm{u}}^{i+1}$ when the $C[i]$ is pulled up to logic "1".

The CBDC-PUF response generation process is divided into three stages, which are the initialization, excitation, and data latching stage. To explain the CBDC-PUF response generation process clearly, $C[n\text{-}1]$ is set as logic "0" and other bits (e.g., $C[0]$, $C[1]$, . . ., $C[n\text{-}2]$) of $C$ are set as logic "1". The CBDC-PUF enters the initialization stage and all the internal nodes are initialized when the *Excite* and *Latch* pull down to logic "0". In this case, $Q_{\mathrm{u}}$ and $Q_{\mathrm{b}}$ rise to logic "1" and the other internal nodes ($Q_{\mathrm{u}}^{0}$, $Q_{\mathrm{b}}^{0}$ and $Q$) are set to logic "0". In the excitation stage, *Excite* is pulled up to logic "1" and *Latch* is remained at the logic "0" at the same time. The logical value of $Q_{\mathrm{u}}$ and $Q_{\mathrm{b}}$ are transmitted to configurable delay elements successively, and the corresponding signal across the configurable delay elements is transmitted or reversed according to the value of its control signal. The gate delay of the buffer part and inverter part of $i$th upper configurable delay element are denoted as $\phi_{\mathrm{u}}^{i}$ and $\psi_{\mathrm{u}}^{i}$, respectively, and the delay of upper and below AND gate are denoted as $\tau_{\mathrm{u}}$ and $\tau_{\mathrm{b}}$, respectively, while the gate delay of the buffer part and inverter part of $i$th below configurable delay element are denoted as $\phi_{\mathrm{b}}^{i}$ and $\psi_{\mathrm{b}}^{i}$. The total delay $\tau_{Q_{\mathrm{u}}}$ along the path $Q_{\mathrm{b}} \rightarrow Q_{\mathrm{u}}$ and the total delay along the path $Q_{\mathrm{u}} \rightarrow Q_{\mathrm{b}}$ can be expressed as

$$\begin{cases} \tau_{Q_{\mathrm{u}}} = \sum_{i=0}^{n-1}((1-C[i])\phi_{\mathrm{u}}^{i} + C[i]\psi_{\mathrm{u}}^{i}) + \tau_{\mathrm{u}} \\ \tau_{Q_{\mathrm{b}}} = \sum_{i=0}^{n-1}((1-C[i])\phi_{\mathrm{b}}^{i} + C[i]\psi_{\mathrm{b}}^{i}) + \tau_{\mathrm{b}} \end{cases} \tag{1}$$

The upper part and lower part of the configurable delay chain are symmetrically distributed by manual placement and the length of the signal wire of two symmetrical paths are ideally identical. So, $\tau_{\mathrm{u}} = \tau_{\mathrm{b}}$, $\phi_{\mathrm{u}}^{i} = \phi_{\mathrm{b}}^{i}$, $\psi_{\mathrm{u}}^{i} = \psi_{\mathrm{b}}^{i}$, and $\tau_{Q_{\mathrm{u}}} = \tau_{Q_{\mathrm{b}}}$. Due to randomness over the manufacturing process, the main parameters of MOS devices, including the channel length, channel width, oxide thickness, and threshold voltage, are deviated from the ideal value. According to the delay calculation approach presented in Reference [51], the delay of two MOS devices in the same wafer varied with each other. Therefore, $\tau_{\mathrm{u}}$, $\phi_{\mathrm{u}}^{i}$, $\psi_{\mathrm{u}}^{i}$, and $\tau_{Q_{\mathrm{u}}}$ are not equal to $\tau_{\mathrm{b}}$, $\phi_{\mathrm{b}}^{i}$, $\psi_{\mathrm{b}}^{i}$, and $\tau_{Q_{\mathrm{b}}}$, respectively. $\Delta\sigma$ is the total delay difference between the upper and below part of the configurable delay chain, which is presented in Equation (2). Where $\Delta\rho$ is the total delay difference between the upper and below configure delay elements, $\Delta\tau$ is the difference between $\tau_{\mathrm{u}}$ and $\tau_{\mathrm{b}}$, and both the $\Delta\rho$ and $\Delta\tau$ are not equal to 0. Concerning the $\phi_{\mathrm{u}}^{i} \neq \phi_{\mathrm{b}}^{i}$, $\psi_{\mathrm{u}}^{i} \neq \psi_{\mathrm{b}}^{i}$, and $\tau_{\mathrm{u}} \neq \tau_{\mathrm{b}}$, thus $\tau_{Q_{\mathrm{u}}} \neq \tau_{Q_{\mathrm{b}}}$ and $\Delta\sigma \neq 0$. As shown in Figure 1, such cross-coupled structure provides a square wave signal flowing through the symmetrical paths. Due to the delay difference between two symmetrical paths, the square wave signals flowing through the symmetrical paths cannot flip simultaneously. The DFF judges which of the two symmetrical paths is faster and outputs the captured data at the negative edge of $C$ as the PUF response.

$$\Delta\sigma = \Delta\rho + \Delta\tau = \tau_{Q_{\mathrm{u}}} - \tau_{Q_{\mathrm{b}}} = \sum_{i=0}^{n-1}((1-C[i])(\phi_{\mathrm{u}}^{i} - \phi_{\mathrm{b}}^{i}) + C[i](\psi_{\mathrm{u}}^{i} - \psi_{\mathrm{b}}^{i})) + \tau_{\mathrm{u}} - \tau_{\mathrm{b}} \tag{2}$$

The proposed PUF enters the data latching stage when both the *Excite* and *Latch* are pulled up to logic "1". The DFF with a negative-edge clock captures a stable state from two possible states. In the data latching stage, the data on the $Q_{\mathrm{u}}$ is loaded into the flip-flop on the falling edge of $Q_{\mathrm{u}}$. If the *Latch* is pulled down to logic "0", the logical value of $Q$ has remained and the falling edge of

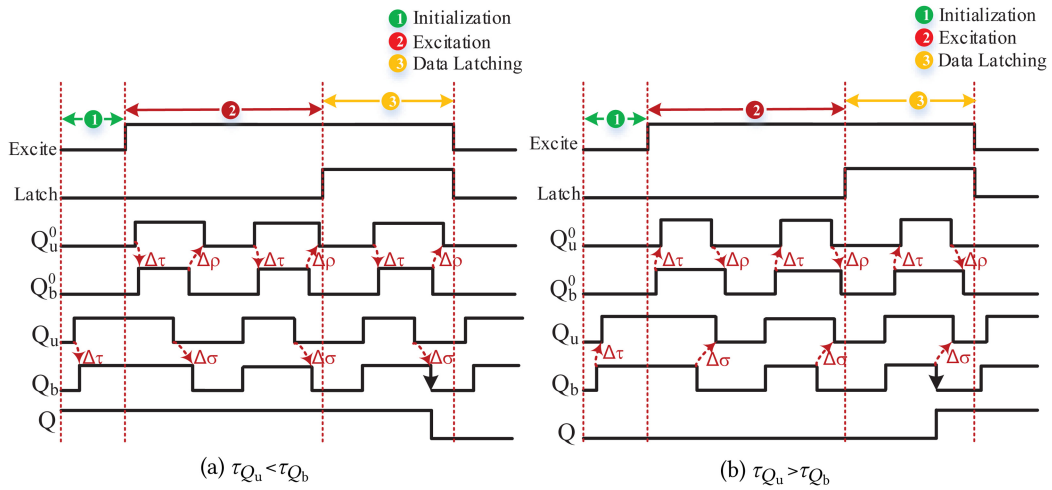



Fig. 2. Timming waveform of proposed PUF.

$Q_u$ is ignored. The output response $Q$ of proposed PUF is a step function of $\Delta\sigma$, which is expressed as

$$Q = sign(\Delta\sigma) = \begin{cases} 0. & \tau_{Q_u} \leq \tau_{Q_b}, \\ 1. & \tau_{Q_u} > \tau_{Q_b}. \end{cases} \tag{3}$$

Figure 2 shows the timing waveform of two possible states of the proposed PUF. In the excitation stage, the logical value of $Q_u$ and $Q_b$ is transferred to the $Q_u^0$ and $Q_b^0$, respectively, when the Excite is pulled up to logic "1", and then the new logical value of $Q_u^0$ and $Q_b^0$ is transmitted or reversed through the configurable delay elements and passed to the $Q_u$ and $Q_b$, respectively. Regarding Figure 2(a), the signal flowing through the upper part of the configurable delay chain is faster than the corresponding below part. In other words, $\tau_b + \tau_{Q_b}$ is greater than $\tau_u + \tau_{Q_u}$. Therefore, $Q_b^0$ and $Q_b$ flip later than $Q_u^0$ and $Q_u$, respectively. Specifically, $Q_b$ and $Q_u$ rise to logic "1" at the same time, while $Q_b$ delays $\Delta\ \sigma$ compared to the $Q_u$ when $Q_u$ goes to logic "0". Similarly, $Q_b^0$ delays $\Delta\ \tau$ at the rising edge and $\Delta\ \rho$ at the falling edge compared with the $Q_u^0$. Figure 2(b) shows another case of the CBDC-PUF. In this case, the total delay of the upper part of the configurable delay chain is greater than the corresponded below part and $\tau_{Q_u} < \tau_{Q_b}$. Thus, $Q_b$ and $Q_b^0$ flip earlier than the $Q_u$ and $Q_u^0$, respectively, and the corresponding delay time is equal to the case in Figure 2(a). In the data latching stage, the logic "0" of $Q_u$ is captured when $\Delta\ \sigma$ is smaller than 0, and thus the response $Q$ of proposed PUF is set as logic "0" in Figure 2(a). For the case in Figure 2(b), the logic "1" of $Q_u$ is captured if $\Delta\ \sigma$ is greater than 0, and thus the response $Q$ of proposed PUF is logic "1". In summary, $\Delta\ \sigma$ caused by process variations is the primary source of randomness, which makes the proposed PUF unique and secure.

### 3.2 Security Analyses

Combining Equations (2) and (3), there exists a linear relationship between $C$ and $Q$. $C$ ranges from 0 to $2^n$, and there exist $2^n$ possible symmetrical paths and possible responses. Once the proposed CBDC-PUF is manufactured, the gate delay of each element in the CBDC-PUF is determined, and $\psi_u^i$, $\psi_b^i$, $\phi_u^i$, $\phi_b^i$, $\tau_u$, and $\tau_b$ are constant. Thus, $\Delta\ \sigma$ can be rewritten as Equation (4). Where $\alpha_0$, $\alpha_1$, ..., $\alpha_{n-1}$ are constant which is related to the fixed gate delay of elements (e.g., $\alpha_i$ is the linear

relationship between $\psi_{\mathrm{u}}^{i}$, $\psi_{\mathrm{b}}^{i}$, $\phi_{\mathrm{u}}^{i}$, and $\phi_{\mathrm{b}}^{i}$).

$$\Delta\sigma = \alpha_0 \times C[0] + \alpha_1 \times C[1] + \cdots + \alpha_{\mathrm{n}-1} \times C[\mathrm{n}-1] + \Delta\tau \propto \mathbf{C} \tag{4}$$

Regarding Equations (4) and (3), $Q$ is linearly correlated with the $\mathbf{C}$. Such a linear relationship between the $Q$ and $\mathbf{C}$ can be established using the machine learning approaches (e.g., **support vector machine** (**SVM**), logistic regression, artificial neural network, and evolutionary strategy) and the PUF response of an arbitrary challenge is easily guessed by the trained model. Therefore, we can conclude that the proposed CBDC-PUF is vulnerable to MA.

The hash function is a mathematical function that converts an arbitrary length of an input value into a fixed length, which is widely applied to password storage, data integrity check, and other information security applications. Several hash algorithms, including the Message Digest, Secure Hash Function, RIPEMD, and Whirlpool, have been proposed over the past decades. Keccak algorithm has efficient performance and good resistance to attacks, which is chose as the new SHA-3 standard by NIST in 2012. No successful attacks have yet been reported on the SHA-3 hash function. Hash function results in substantially different hash values for two messages that differ by even a single bit of data. Based on that, a hash function can increase the complexity of the challenge to response mapping in a strong PUF. Several works have demonstrated that the hash function can obscure the relationship between the challenges and responses and further improve the anti-modeling attack ability of PUF [52–54]. Therefore, a secure hash algorithm is further used to eliminate the linear relationship between $Q$ and $\mathbf{C}$ in this article. Equation (3) can be re-written as Equation (5). Where non-linear function $f$ implements the secure hash algorithm. As shown in Equation (5), a nonlinear relationship exists between $Q'$ and $\mathbf{C}$ which shows that the proposed CBDC-PUF is resilient to MA approaches.

$$Q' = sign(\Delta\sigma) = sign\left(\sum_{i=0}^{\mathrm{n}-1} \alpha_i f(\mathbf{C}) + \Delta\tau\right) \propto f(\mathbf{C}) \tag{5}$$

## 4 EXPERIMENTAL RESULTS AND ANALYSES

### 4.1 Experiment Setup

For a CBDC-PUF, the width of the control signal is set as 16-bit. There exist $2^{16}$ pairs of symmetric paths for a CBDC-PUF and thus the number of CRPs is 65536. A total of 128 CBDC-PUFs are arranged in a matrix and thus the CRPs space of the PUF framework is $65536^{128}$. Furthermore, a secure hash algorithm-3 (SHA-3) is used to generate 2048-bit hash data that configures $65536^{128}$ pairs of symmetric paths of the PUF framework. To improve the robustness of the proposed PUF, several primitives of FPGA, including the LUT2, NAND2, and FDCE_1, are used to build the structure of CBDC-PUF. Besides, the placement of CBDC-PUF is adjusted and fixed using the FPGA Editor, and the routing of two differential signal paths is modified to be approximately equal. The fixed structure of CBDC-PUF is saved as a hard macro and reused to form a 128-bit PUF response. The proposed PUF occupies 256 NAND2s, 8192 LUT2s, and 128 FDCE_1s, while the total design requires 4406 registers, 9544 LUTs, 4568 slices, 4 bonded IOBs, and 1 BUFG. The utilization of the number of slice registers is 3%, the utilization of the number of slice LUTs is 15%, the utilization of the bonded IOBs is 1%, and the utilization of the number of the BUFG is 3%. Moreover, we compare the FPGA utilization results of the proposed design with the several works (FFXOR [32] and 4-XOR APUF [23]) in Table 1. We use the Xilinx XPower Analyzer to calculate the power consumption of total design. The dynamic supply power is 0.427 W, and the quiescent supply power is 0.089 W, and the total supply power is 0.516 W. We also calculate the computation latency of the total design, and the computation latency is 10.627ns and the maximum frequency is 92.799 MHz.

Table 1. The FPGA Resource Usage Results of the PUFs

| PUF Type | FPGA resource usage | | | | |
|---|---|---|---|---|---|
| | NAND2 | MUX2 | LUT2 | XOR2 | DFF |
| FFXOR [32] | - | 16384 | - | 896 | 128 |
| 4-XOR APUF [23] | - | 65536 | - | 384 | 512 |
| CDBC-PUF | 256 | - | 8192 | - | 128 |

In the actual situation, PUFs are subject to variations of environmental conditions, and the responses have several error bits under the same challenge. However, the PUF-based application scenarios require an error-free response as the authentication password or cryptographic key. So, an error correction code should be considered to correct several bit errors from the PUF response. Considering the hardware complexity and error-correcting capability, an error correction algorithm based on the BCH code is adopted and used to solve the bit-error problem of PUF response. In this article, the BCH code is characterized as (31, 16, 3) and 3-bit errors can be corrected from the 16-bit response.

To evaluate the performance of the proposed PUF, an FPGA-based PUF test platform is built to acquire the CRPs under varying test conditions. The total project of the PUF framework runs on the RISC-V evaluation board (Nuclei MCU200T) [55], and the corresponding FPGA is Xilinx XC7A200T-2FGG484. A serial communication interface establishes a communication link between the PC and FPGA-based platform. Besides, an automatic PUF response acquisition tool based on the LabVIEW is built to obtain responses with the same challenge multiple times and analyze its performance results (e.g., robustness, uniqueness, and randomness).

### 4.2 Performance Analysis

Process variations, including inter-die variations and intra-die variations, are the sources of PUFs. In this article, we use an FPGA to validate the efficacy of PUF and only exploit the intra-die variations to generate the PUF response. We change the location of the occupied FPGA primitives and design 10 macros to emulate 10 PUF implementations. Randomness, stability, and uniqueness are the main performance metrics of the PUF, and the detailed calculation methods are presented in [56]. A total of 50000 random challenges are generated and applied to the proposed PUF. Figure 3 shows the randomness results of 50000 responses. As shown in Figure 3, the randomness ranges from 27.3% to 61.7% and more than 81.9 percent of responses fall within the range of 40% to 60%. For the randomness quantification approaches, the much closer to 50% the results are, the more uniformly the logic “0” and logic “1” distribute, and the better the randomness of results is. To ensure the unpredictability characteristic of the PUF, the randomness results of 11202 CRPs range from 48.4% to 51.6%, which are chosen and used as the candidate for security fields. Figure 4 shows the distribution of logic “0” and “1” of those responses. Where the black and white filled box represent the logic “0” and “1”, respectively, the horizontal axis of this figure is the value of each response bit, and the vertical axis represents the PUF response value of each bit. Regarding the Figure 4, it can be observed that the logic “0” and logic “1” are uniformly distributed. More specifically, the averaged distribution of logic “0” and logic “1” are 49.4% and 50.6%, which is very close to the ideal value 50%.

Furthermore, the NIST statistical test suite is used to further quantify the randomness of PUF responses. Notice that the NIST SP 800-22 includes 15 test items and each test item outputs the $p$-value to quantify the level of randomness. A larger $p$-value indicates a higher confidence level of randomness. If the $p$-value of a data exceeds 0.01, this data is considered as random with 99% confidence [57]. In this article, nine test items are implemented and the corresponding $p$-value results are shown in Table 2. Where the block frequency, the longest run of ones, discrete Fourier

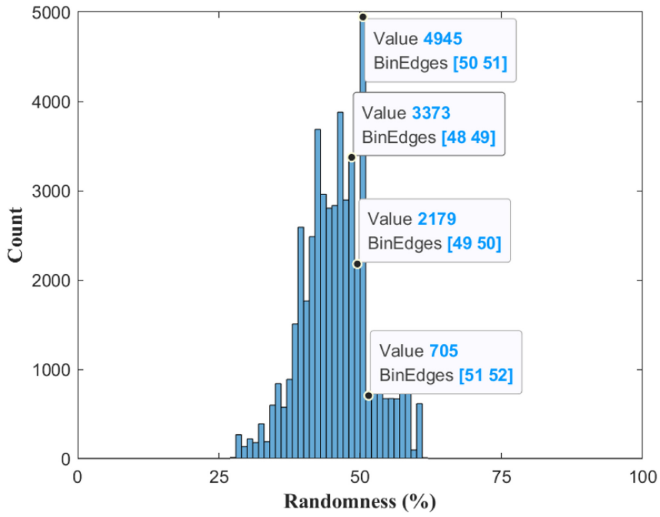



Fig. 3. Randomness results of PUF.

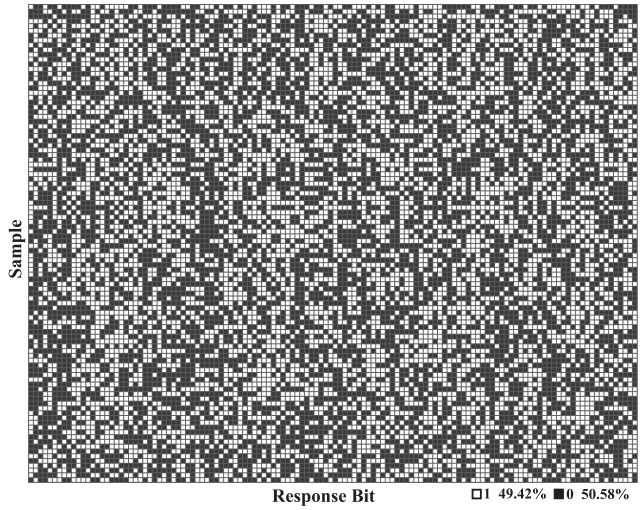



Fig. 4. The distribution of logic "0" and "1" of PUF responses.

transform, nonperiodic template matchings, approximate entropy, and cumulative sums are denoted as BF, LRO, FFT, NOT, AppEn, and CumSums, respectively. The width of response is 128-bit and thus the bitstream length (denoted as $n$) is 128, which makes the remaining 6 test items not executed. There exist 11202 responses and the number of bitstreams (denoted as $m$) is 11202. The fourth and fifth columns are the averaged $p$-value (denoted as Avg-p) and ratio (denoted as RPT) of pass number to total number, respectively. As shown in Table 2, the $p$-values of all the test items are greater than 0.01, the responses can pass the Frequency, BF, LRO, and CumSums successfully, and there exist several failure responses during the Runs, FFT, NOT, Serial, and AppEn tests. Although a small portion of responses fails to pass the test items, more than 92.7% responses pass the test items. In summary, we can conclude that the responses of the proposed PUF are uniformly random.

Table 2. The NIST Test Results of PUF Responses

| NIST tests | Bitstream length (n) | Number of bitstreams | Averaged *p*-value | Ratio of pass number (%) |
|---|---|---|---|---|
| Frequency | 128 | 11202 | 0.85 | 100 |
| BF | 128 (M = 128) | 11202 | 0.85 | 100 |
| Runs | 128 | 11202 | 0.47 | 99.6 |
| LRO | 128 | 11202 | 0.47 | 100 |
| FFT | 128 | 11202 | 0.50 | 98.4 |
| NOT | 128 (m = 5) | 11202 | 0.64 | 92.7 |
| Serial | 128 (m = 4) | 11202 | 0.48 | 96.4 |
| AppEn | 128 (m = 4) | 11202 | 0.46 | 96.4 |
| CumSums | 128 | 11202 | 0.85 | 100 |

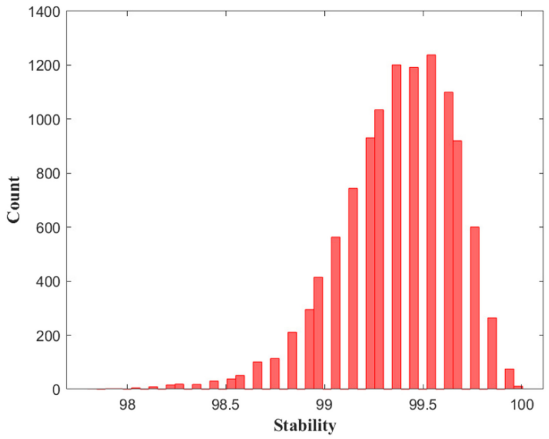



Fig. 5. The stability distribution histogram of total responses.

In addition, stability is used to evaluate the ability of PUF to produce the same responses under noisy operating conditions. Where the stability value is the ratio between the error-free bits and total bits of a response. Figure 5 shows the stability distribution of responses. As shown in Figure 5, all the stability values range from 97.8% to 100%, which is close to the ideal value (100%). Besides, 95% stability values of responses are greater than 98.8%, and the mean value of stability is 99.4%.

The unstable bits are analyzed from 1000 repeated responses and the corresponding map results are shown in Figure 6. The whole map consists of 128 small blocks and each block represents one bit of responses, which are arranged from left to right. More specifically, the first row corresponds to the lower 16-bit of response and the leftmost block of the first row is the lowest bit of response. The value floating at the block is the sum of 1000 responses for this bit. If all the responses of this bit are logic "1", the corresponding cumulative value ($C_V$) is 1000; conversely, $C_V$ is 0 when all the responses of this bit are logic "0". Otherwise, $C_V$ is not equal to 1000 or 0 when there exist several errors in PUF responses, and the unstable value is equal to $C_V$ if $C_V$ is smaller than 500 or 1000-$C_V$ if $C_V$ is greater than 500. For the map, the blocks with a red square are used to highlight the unstable bits of responses. Figure 6 clearly shows that 11 unstable bits exist, the unstable bits are disorderly distributed and the corresponding $C_V$ ranges from 1 to 999. More specifically, the minimum and maximum error values are 1 and 209, respectively.

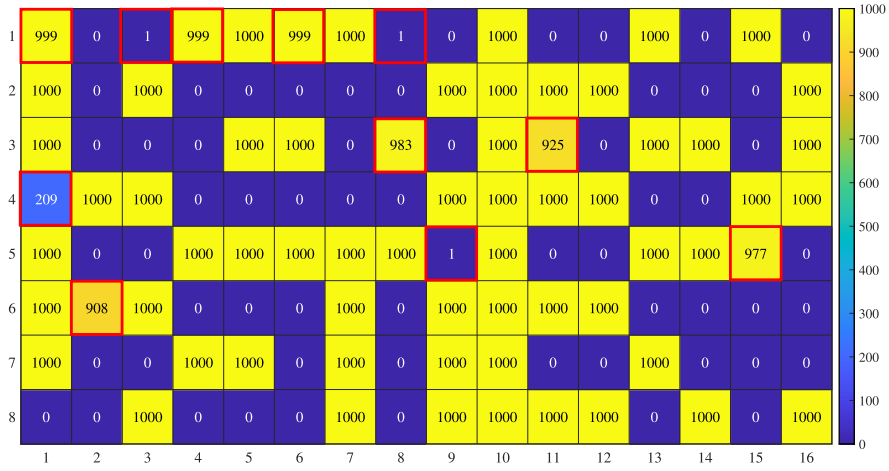



Fig. 6. Unstable bit map results of responses.

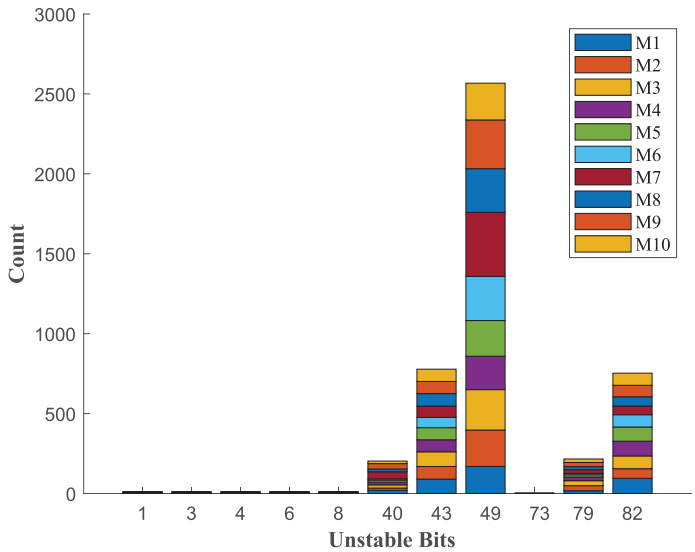



Fig. 7. Unstable bits distribution of responses under noisy operating conditions.

The PUFs are tested 10 times repeatedly under different situations and the unstable bit count results are shown in Figure 7. A total of 10 test results are denoted as $M_1$, $M_2$, . . ., $M_{10}$. Regarding Figure 7, 11 unstable bits are 1, 3, 4, 6, 8, 40, 43, 49, 73, 79, and 82. Most bit errors are concentrated on the 40th, 43th, 49th, 79th, and 82th bit, and the ratio of errors in these five unstable bits to total errors is 98.9%. Therefore, we assume that the 40th, 43th, 49th, 79th, and 82th bits are the unstable bit of responses. Besides, the remaining bits have only one error in 1000 repeated tests. In this case, such bit errors are mainly caused by the noisy operating conditions at runtime. Consequently, the bit error problem should be considered and an error correction code is needed to correct the unstable bits.

Uniqueness is an indication that a PUF differs from other PUFs, and the inter-distance value of 3 PUFs with 10000 challenges is used to quantify the uniqueness metric of PUFs. Figure 8 shows the

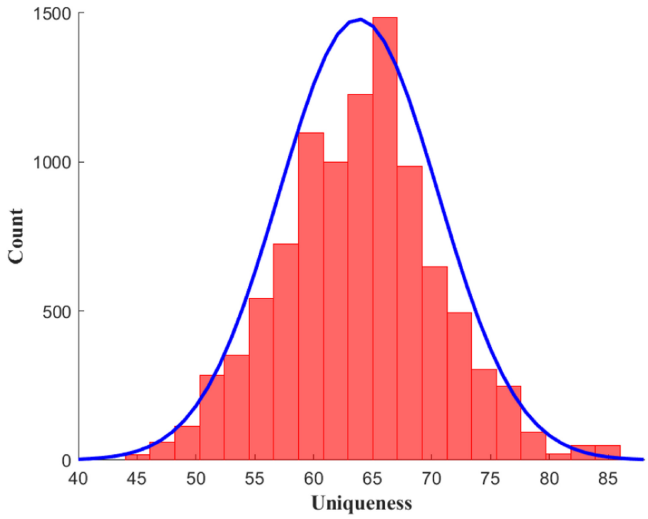



Fig. 8. Uniqueness distribution of PUF responses.

distribution of uniqueness results. Regarding Figure 8, the uniqueness ranges from 44 to 85, and the mean and standard deviation of uniqueness are 63.8 and 6.7, which are very close to the ideal value of 64. Therefore, each PUF has a unique response and it is infeasible to produce the same responses of two PUF instances by applying the same challenge. In summary, the proposed PUF has robust, random, and unique properties, which presents an appealing option for a cost-sensitive and high-security embedded system.

### 4.3 MA-Resilience Analysis

To evaluate the MA-resilience ability of the proposed PUF, several popular machine learning algorithms, including the SVM, **generalized regression neural network** (**GRNN**), and **radial basis function neural network** (**RBFNN**), are used to establish the mathematical model with a set of CRPs and predict the PUF responses of arbitrary challenges. However, other empirical machine learning methods (e.g., logistic regression, artificial neural network, and naive Bayes) presented in References [26, 29, 32] are not used to assess the MA-resistance of proposed PUF because those methods have a poor training efficacy. In this article, the **mean squared error** (**MSE**) is used to represent the quality of the training model. The smaller the MSE is, the more the training model matches the real relationship between the challenges and responses. Figure 9 shows the trend of training MSE of RBFNN. The horizontal axis is the epoch that represents the iteration process of model parameters in the machine learning algorithms, and the vertical axis is the value of MSE.

Regarding Figure 9, the training MSE decreases dramatically with the increasing of epochs, and the training MSE reaches 6.67 $e^{-26}$ when the epoch is equal to 700. Besides, 70% of CRPs are used to train the model and the training accuracy is calculated to further validate the efficacy of the training model. The training accuracy with different numbers of CRPs is shown in Table 3. It is clear from Table 3 that all the training accuracy of three training models (SVM, GRNN, and RBFNN) are approximately equal to the ideal value (100%). In summary, the training model scales well with the relationship of CPRs.

The training model is used to evaluate the resistance against MA and the prediction results are shown in Table 3. The ideal prediction accuracy of high MA-resistance PUF is 50%, which shows that the machine learning techniques are similar to random guessing. As shown in Table 3, the

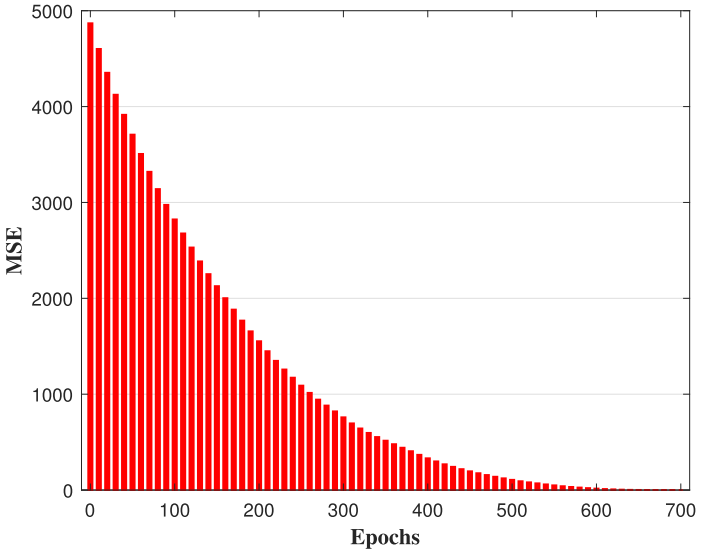



Fig. 9. The trend of training MSE for training RBFNN model.

Table 3. Training and Prediction Results of Three Machine Learning Algorithms

| Performance \ Number of CRPs | | 100 | 500 | 1000 | 2000 | 5000 | 10000 | Averaged |
|---|---|---|---|---|---|---|---|---|
| Training Accuracy | SVM | 100% | 100% | 100% | 100% | 100% | 99.9% | 99.9% |
| | RBFNN | 100% | 100% | 100% | 100% | 99.5% | 99.8% | 99.8% |
| | GRNN | 100% | 100% | 100% | 100% | 100% | 100% | 100% |
| Prediction Accuracy | SVM | 49.2% | 49.2% | 51.5% | 50.5% | 50.6% | 50.5% | 50.3% |
| | RBFNN | 52.5% | 51.7% | 49.1% | 49.8% | 51.9% | 50.5% | 50.9% |
| | GRNN | 51.1% | 50.5% | 50.2% | 50.9% | 51.1% | 50.9% | 50.8% |

prediction accuracy of the three models ranges from 49% to 51%, which is close to the ideal value (50%). Moreover, the trend of prediction accuracy remains stable and there is no increasing trend with the increasing number of CRPs. More specifically, the averaged prediction accuracy of SVM, GRNN, and RBFNN, respectively, are 50%, 51%, and 50.5%, which proves that the proposed PUF can resist machine learning techniques. The prediction process of the established model is similar to random guessing, and machine learning algorithms fail to build an efficient PUF model. Therefore, we can conclude that the proposed PUF is resilient to MA.

### 4.4 Comparison with Prior Works

In this subsection, we present the performance comparison of proposed PUF with the state-of-the-art studies. The comparison results are listed in Table 4. For the proposed PUF, two symmetrical paths of the proposed PUF are switched with the challenges. By using this path configurable strategy, the proposed PUF can be configured as various forms and output a large set of responses. As mentioned in Section 4.1, the CRP space size of the proposed PUF is $2^{2048}$, which is significantly larger than the remaining works. Moreover, both the randomness and uniqueness of all the methods are close to the ideal value (e.g., 50%). Although the response of the proposed PUF may flip in a noisy environment, all the unstable bits can be screened out

Table 4. Performance Comparison

| PUFs / Performance | MPUF [58] | 5-4 DAPUF [27] | SD-PUF [28] | A-PUF [25] | CT-PUF [29] | Proposed PUF |
|---|---|---|---|---|---|---|
| Configurable | Yes | Yes | No | Yes | Yes | Yes |
| Bit length | - | 64 | - | 128 | 128 | 128 |
| CRP space size | - | - | $2^{12}$ | $2^{128}$ | $2^{256}$ | $2^{2048}$ |
| Randomness | close to 50%* | 49.9% | 49.9% | - | close to 50%* | 49.6% |
| Reliability | 99.6% | 79%~88% | 100% | 100% | close to 100%* | 100% |
| Uniqueness | 49.8% | 44.2% | 49.9% | 49.9% | close to 50%* | 49.8% |
| Prediction accuracy | 53.8% | 50%~57% | 60% | close to 50%* | 60% | 50% |
| MA-resilience ability | Yes | Yes | Yes | Yes | Yes | Yes |
| Bit-instability correction | No | No | No | Yes | No | Yes |

*The exact value is not given in this article. - The value is not given in this article.

using the bit-instability correction mechanisms. In summary, the performance of the proposed PUF is comparable with the existing methods. Besides, the prediction accuracy of proposed PUF fluctuates around 50%, which is comparable with that in Reference [25] and superior to that in the other works (References [27–29, 58]). Thus, the proposed PUF demonstrates a high MA-resilience ability than the state-of-the-art methods. Considering the trend of prediction accuracy for existing PUFs, the traditional machine learning algorithms are hard to establish the PUF model with a set of CRPs, and the machine learning-based prediction is equivalent to random guessing. Therefore, the proposed PUF is more resilient to MA than other PUFs in the literature. On the whole, the advantages of the proposed PUF, including the CRP space size, performance, and MA-resilience, are obvious. Consequently, we can conclude that the proposed PUF is superior to the existing works.

## 5 CONCLUSION

In this article, a CBDC-PUF is proposed, which exploits the intrinsic delay differences between two symmetrical paths to produce a large set of high responses. Furthermore, a cryptographic hash function is presented to obfuscate the linear relationship between the challenges and responses, which improves the resistance against MA. Moreover, a bit-instability correction mechanism is used to correct unstable bits under noisy operating conditions, which enhances the robustness of the proposed PUF. Experimental results show that the main performance and MA-resilience of the proposed PUF are closer to the ideal value. The inter-die variations are not considered in this article, and we will further purchase more FPGAs to further validate the efficacy of PUFs. In the future, several PUF-based security solutions are explored to resist software and hardware attacks and ensure the integrity and trustworthiness of embedded systems.

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